رواد مصر الرقمية

STARTUP SUCCESS ANALYSIS

Funding & Investment Outcomes | 5,000 Startups | 7 Industries

Technical Instructor

Amal Mahmoud

Project Team

Khadeja Fekry · Filopater Maged · Omnia Yousry
Carol Hany · Seif Taha · Mohammed Emad

Tools: Excel · Power BI · Tableau · Python · SQL

Audience: Investors & Venture Capital Stakeholders

The Team

Six analysts, one technical instructor, and a shared obsession with turning raw startup data into something investors can actually use.

Name	Contribution
Khadeja Fekry	Excel Dashboard Development, Python Analytics & Automation, Project Documentation
Filopater Maged	Excel Dashboard Development, Python Analytics & Automation
Mohammed Emad	Data Modeling & Star Schema Design, Pivot Table Analysis
Omnia Yousry	Power BI Dashboard Development & Interactive Reporting
Seif Taha	MySQL Database Engineering & Query Development
Carol Hany	Tableau Visualization & Interactive Dashboard Design

Technical Instructor: Amal Mahmoud

Tools & Technologies: Microsoft Excel · Power BI · Tableau · Python · MySQL

1. Introduction

Every year, thousands of startups compete for a limited pool of investor capital — and most of them won't survive long enough to see a second funding round. What makes the ones that do succeed actually different? That's the question this project was built to answer.

Our team analyzed 5,000 startups across seven industries — AI/ML, Education, Tech, Healthcare, Logistics, E-Commerce, and Finance — using a full data analytics stack. The data was structured into a star schema in MySQL, explored using Python, modelled in Excel, and visualized in both Power BI and Tableau. The result is a cross-sector, evidence-backed view of what drives funding success.

This report documents the project's Introduction, Objective, and Challenges — giving investors and stakeholders the full picture of what was built, why, and what we found.

The Dataset at a Glance

Metric	Value	Source
Total startups analyzed	5,000	Raw-Data sheet (CSV + Excel)
Industries covered	7	AI/ML, Education, Tech, Healthcare, Logistics, E-Commerce, Finance
Total funding across all startups	$124.9 billion	Pivot Table — Industry Funding
Total revenue across all startups	$249.6 billion	Pivot Table — Market & Growth
Average funding per startup	$24.97 million	Pivot Table — Avg. Funding per Industry
Average revenue per startup	$49.92 million	Pivot Table — Avg. Revenue
Average founder experience	14.74 years	Efficiency & HR analysis
Average marketing ROI	275.8x	Pivot Table — Marketing Efficiency
Startup cohort period	2012 – 2026	Dim_Date table

Industry Breakdown

The cohort is well-distributed across all seven industries, with AI/ML slightly leading in count (743 startups), closely followed by Education (741) and Tech (723). Each sector has its own funding dynamics — and the numbers tell some surprising stories.

Industry	# Startups	Total Funding Raised	Total Revenue	Avg. Revenue/Startup
AI/ML	743	$18.68B	$37.31B (net profit)	$50.72M
Education	741	$18.78B	$35.40B (net profit)	$48.29M
Tech	723	$18.42B	$35.47B (net profit)	$49.55M
Healthcare	722	$18.25B	$36.86B (net profit)	$51.55M
Logistics	721	$17.34B	$36.26B (net profit)	$50.80M
E-Commerce	678	$16.97B	$33.50B (net profit)	$49.91M
Finance	672	$16.42B	$32.30B (net profit)	$48.56M
TOTAL	5,000	$124.87B	$247.10B (combined net profit)	$49.92M

Healthcare leads on average revenue per startup at $51.55M despite not being the most funded sector — a sign of strong capital efficiency. AI/ML, meanwhile, generates the highest total net profit ($37.3B), confirming its status as the sector attracting both the most startups and the most productive investment.

2. Objective

The core objective of this project is straightforward: to give investors a clear, data-driven answer to which startups are worth backing — and why. We wanted to move beyond gut feel and build an evidence base that covers funding, profitability, operational efficiency, and growth patterns across seven industries.

To make that happen, the team structured the analysis around three main questions:

- Where is investor capital actually generating returns — and which industries are the most capital-efficient?
- How efficiently are startups managing their burn, and what does that mean for their funding runway?
- What patterns in revenue, marketing spend, and employee productivity distinguish high-performing startups from struggling ones?

The KPIs We Tracked

The analysis in our Excel workbook centered on the following KPIs — drawn directly from the pivot tables, financial analysis sheets, and the star schema data model built in MySQL:

KPI	What It Measures	Real Finding from Our Data
Funding Amount	Total capital raised per startup	Avg. $24.97M per startup; Education sector leads at $25.35M avg.
Net Profit	Revenue minus costs — the real measure of viability	Hybrid business model leads at $83.15B total; B2B: $82.25B; B2C: $81.69B
Burn Rate & Burn Multiple	Monthly cash consumption and spend efficiency	Veteran startups burn less efficiently (0.204) vs. New startups (0.038)
Revenue Per Employee	Productivity of each hire — HR efficiency signal	AI/ML leads at $445,607/employee; Tech is lowest at $255,162/employee
Marketing ROI	Return generated per dollar spent on marketing	Large market startups lead at 305.3x; Small market: 269.7x; Medium: 250.3x

Customer Retention Rate	Percentage of customers retained across the period	Consistent across industries — avg. 50.2%, ranging from 49% to 51.2%
Funding per Founder	Capital share and responsibility per founding team member	Avg. $24.97M total / avg. 4,151 total founders across 5,000 startups
Revenue Growth by Year	How total revenue across the cohort shifts over time	Peak at $18.33B in 2024; 2026 drops to $15.27B — a recency effect in the data

Business Model Performance

Key Insight: The Hybrid business model generates the highest total net profit ($83.15B), slightly ahead of B2B ($82.25B) and B2C ($81.69B). For startups targeting fast profitability, a hybrid revenue approach appears to be the most effective structural choice in this dataset.

What the Analysis Was Designed to Deliver

Beyond the KPIs, the project was set up to produce four concrete, investor-ready outputs:

- A star schema data model (MySQL) structuring all 5,000 startups across industry, market size, business model, and founding year dimensions.
- Python-driven analysis automating KPI computation and pattern detection across the full dataset.
- Excel financial models and pivot tables surfacing sector-level and cohort-level comparisons — including the top 10 startups by revenue, industry profitability rankings, and burn efficiency by startup maturity.
- Interactive dashboards in Power BI and Tableau enabling investors to filter by industry, market size, business model, and founding period to drill into the metrics that matter most to them.

3. Challenges

Running a full-stack data project — SQL to Python to Excel to two BI tools — across a team of six people is genuinely complex. Here's an honest account of what made this hard and what we learned from it.

Time Management

The biggest challenge was time, and it showed up at every phase. Coordinating six analysts working across five different tools — each with their own dependencies — meant that delays in one layer rippled into others. When the MySQL schema needed an adjustment, the Python scripts that read from it needed updates too, which then cascaded into the Excel models and the Power BI connections.

We underestimated how long the multi-tool integration work would take. SQL to Python hand-offs, Python outputs feeding Power BI, Excel pivot tables cross-referencing the same base data — keeping all of that consistent required far more coordination time than pure analysis time.

The documentation phase — this report — also came at the end, under time pressure. Getting the tone right for an investor audience while accurately reflecting what we actually built took real effort.

How We Handled It

What we did: Prioritized the three highest-impact deliverables first (data model, KPI computation, dashboards),
and set internal gates between phases. No dashboard work started until the SQL schema and Python outputs were validated.
Documentation was scoped and outlined early, even if the writing came last.

Cross-Tool Consistency

With five tools all reading from or referencing the same underlying data, keeping numbers consistent was a persistent challenge. A figure in the Tableau dashboard needed to match the corresponding Excel pivot table, which needed to match the Python output, which needed to match the MySQL query. Any change upstream required a cascade of checks.

We established a single source of truth early — the MySQL database — and required that all tools pull from validated base queries rather than local copies of the data. This reduced inconsistencies significantly, though it required discipline to maintain throughout the project.

Defining What 'Success' Actually Means

Across seven industries with fundamentally different business models, revenue structures, and capital requirements, there is no single definition of a 'successful' startup. A $5M raise is transformative in Education; it barely registers in AI/ML. A burn multiple of 0.04 is excellent; 0.2 could signal a problem or simply reflect a growth-stage investment strategy.

We addressed this by building sector-relative benchmarks rather than universal thresholds — which is why the pivot tables are structured by industry, business model, and market size rather than producing a single ranking. The dashboards give investors the filter controls to apply their own definition of success based on their portfolio thesis.

Challenge	Impact	Resolution
Time management across 6 analysts & 5 tools	High	Phased delivery gates; tool dependencies mapped upfront
Cross-tool data consistency	High	MySQL as single source of truth; validated at each hand-off
Defining 'success' across 7 different industries	Medium	Sector-relative benchmarks; investor-controlled dashboard filters
Coordinating analysis vs. documentation timelines	Medium	Documentation outlined early; writing executed in final sprint
Business model comparison (B2B vs B2C vs Hybrid)	Low-Medium	Segmented all KPIs by business model in pivot tables and dashboards

What We'd Do Differently

If we were starting again, we'd automate the SQL-to-Python-to-dashboard pipeline from day one rather than building it sequentially. We'd also allocate documentation time from the start as a dedicated workstream — not as a final deliverable crammed into the last week. And we'd build the star schema validation checks before loading any data, rather than catching mismatches mid-analysis.

These are lessons that came from doing the work, not from reading about it. The analysis is solid. The process got us there — just with some pressure points we'd smooth out next time.

Document Status & Deliverables

This document covers Sections 1–3 of the full Startup Success Analysis Report.
The complete deliverable package includes the MySQL schema, Python analysis scripts,
Excel workbook with all pivot tables and financial models, Power BI dashboard, and Tableau dashboard.

For questions, contact any team member or reach out through your project coordinator.